Week 1, Day 1 — Finding the Right Model

Embeddings & Retrieval Foundations · Clinical AI Internship

Today's job was simple to state and harder to prove: pick the right embedding model before building anything on top of it. I didn't have a real clinical dataset yet, so I built a small synthetic medical corpus — 26 fake patient notes, discharge summaries, radiology reports, and nursing notes — enough to run a real test without waiting on data access approvals.

The Core Question

Medical text is full of shorthand a general-purpose AI model has never been trained to understand. Does a doctor's note that says “Pt c/o SOB, hx of DVT” actually get recognized as meaning the same thing as “patient complains of shortness of breath, history of deep vein thrombosis”? That was the thing to find out before building a search engine on top of it.

Round 1: Bare Abbreviations

First test was blunt — just feed the model pairs like “MI” and “myocardial infarction” and see if it agreed they meant the same thing. The general-purpose model (MiniLM) scored this pair at 0.092 — essentially unrelated in its eyes. The medical-specific model (PubMedBERT) scored it at 0.928. That's not a small gap, that's the difference between a model that has read medical literature and one that hasn't.

But then the hard-negative test caught something interesting. “MI” and “Michigan” should score low, since they mean nothing alike. PubMedBERT scored that pair 0.940 — just as high as the real synonym pair. Turns out very short, bare-word inputs confuse even a good model; this is a known quirk called anisotropy, where isolated short phrases all get pulled toward a similar score regardless of actual meaning.

Round 2: Full Sentences

Fixed the test by comparing full clinical sentences instead of bare words — much closer to how the real system will actually be used, since it'll be comparing whole notes, not isolated terms.

PubMedBERT's synonym sentences averaged around 0.94 similarity, MiniLM's averaged around 0.40. The gap held up. PubMedBERT's negative sentences were still somewhat high (~0.82), confirming the anisotropy issue doesn't fully disappear even at sentence level — but the important thing is the relative separation, not the absolute number.

Round 3: Does It Actually Retrieve the Right Document?

The test that actually matters isn't “do these two sentences look similar” — it's “when someone searches a real question, does the correct document actually come back near the top, out of all 26 notes?” Ran 10 sample clinical queries against the full corpus for both models and checked where the correct answer landed in the ranking.

Model	MRR	Hit@5
General (MiniLM)	0.908	0.900
Medical (PubMedBERT)	0.933	1.000

PubMedBERT won on both counts, but the number that actually told the story was buried in one query: “patient with shortness of breath and history of blood clot.” The correct note used shorthand only (“SOB,” “hx,” “DVT”) with zero literal word overlap with the query. MiniLM ranked it #13 — completely buried, a real clinician would never find it. PubMedBERT ranked it #3 — not perfect, but actually visible.

Every other query in the set had strong literal word overlap between the query and the correct document, so both models nailed rank #1 — those queries didn't really test anything. The one hard query was the only one that mattered, and it's exactly the kind of query a real clinical search system will face constantly: paraphrased, abbreviated, informally worded.

The Takeaway

> “Both models perform well on literal-overlap queries. On the one synonym-dependent query, PubMedBERT ranked the correct document dramatically higher (#3 vs #13) than the general model — confirming domain-specific embeddings meaningfully improve retrieval on paraphrased and shorthand clinical queries, which are common in real EHR text.”

Decision made: PubMedBERT is the model going into the rest of Week 1 — the BM25 baseline tomorrow, the hybrid search on Wednesday, and re-ranking to close out the week. The synthetic eval set also has a real limitation worth flagging going forward: it's too easy overall, and some queries genuinely have more than one correct answer, which the current scoring doesn't account for.

What's Next

Day 2 builds the counterpart to today's semantic search: BM25, a plain keyword-matching search with no understanding of meaning at all. It exists because keyword search is more reliable for exact matches — drug names, dosages, codes — things a meaning-based model can sometimes fuzz over. Both approaches get tested side by side, and by Thursday they'll be combined into one hybrid pipeline.